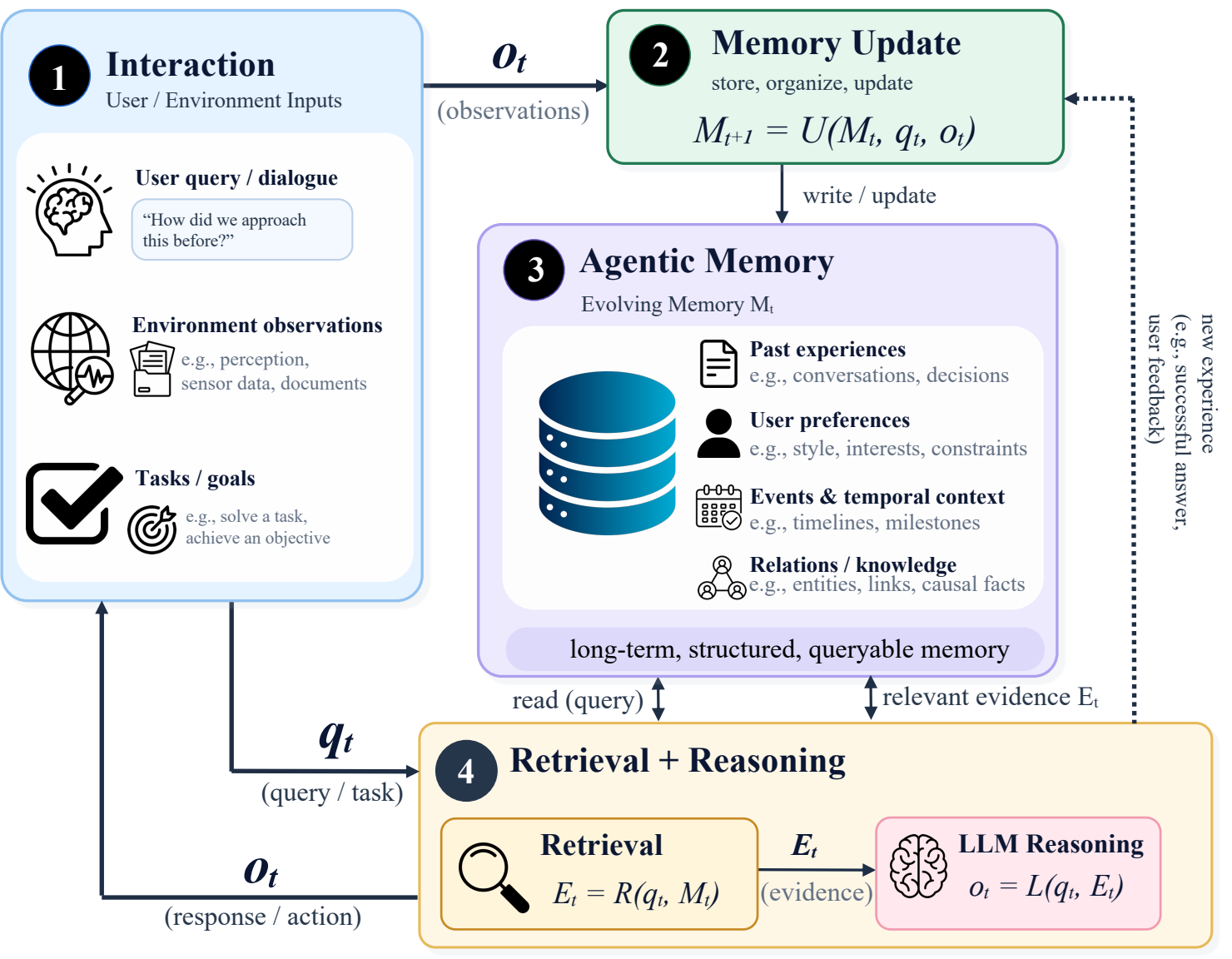

1
Interaction
User / Environment Inputs
User query / dialogue
"How did we approach this before?"
Environment observations
e.g., perception, sensor data, documents
Tasks / goals
e.g., solve a task, achieve an objective
o_t
(observations)
2
Memory Update
store, organize, update
$M_{t+1} = U(M_t, q_t, o_t)$
write / update
3
Agentic Memory
Evolving Memory M_t
Past experiences
e.g., conversations, decisions
User preferences
e.g., style, interests, constraints
Events & temporal context
e.g., timelines, milestones
Relations / knowledge
e.g., entities, links, causal facts
long-term, structured, queryable memory
new experience
(e.g., successful answer, user feedback)
read (query)
relevant evidence E_t
q_t
(query / task)
4
Retrieval + Reasoning
Retrieval
$E_t = R(q_t, M_t)$
E_t
(evidence)
LLM Reasoning
$o_t = L(q_t, E_t)$
o_t
(response / action)